Spring 2026 Analytics II Team Project Final Report

Group 1

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Abstract

Manufacturing defect detection is critical for maintaining product quality, reliability, and safety in industrial systems. In manufacturing, undetected defective components can lead to equipment failures, increased maintenance costs, scrap, rework, and potential safety risks. This project applies machine learning techniques to predict defective parts using historical manufacturing test data.

The dataset contains 229,999 observations, including 850 duplicate records, 0 missing values, and 30 numerical predictor variables representing sensor readings or process measurements. The target variable is a binary defect label, where Class 1 represents defective parts. Only 458 samples are defective, giving a defect rate of 0.1991%. This creates a severe imbalanced classification problem, where traditional metrics such as accuracy can be misleading because a model may achieve high accuracy while failing to identify rare defects.

To address this challenge, this project first compares several machine learning models. Because the defective class is extremely rare, each model is tested with different imbalance-handling strategies, including oversampling, undersampling, SMOTE, and class-weighted learning. Different data splits are also compared to check whether performance remains stable. Finally, models are ranked using a custom FN & FP Score based on false negative and false positive rates, since overall accuracy can appear high even when rare defects are missed.

1 Introduction

Manufacturing defect detection plays an important role in improving product quality, reducing waste, and preventing costly failures. In manufacturing, defective components can contribute to equipment damage, increased maintenance, production delays, customer dissatisfaction, and potential safety risks if they are not detected early. As manufacturing systems collect more process and test data, machine learning provides an opportunity to identify defect patterns that may not be obvious through manual inspection alone.

This project uses historical manufacturing data to predict whether a manufactured part is defective. Of the full dataset, only 0.1991% of samples are labeled as defective. This makes the problem a severe imbalanced classification task.

The main challenge is that the defective class is extremely rare. This project first compares several machine learning models on this unbalanced data. Because the defective class is extremely rare, each model is tested with different imbalance-handling strategies. Different data splits are also compared to check whether performance remains stable. Finally, models are ranked using a custom FN & FP Score based on false negative and false positive rates, since overall accuracy can appear high even when rare defects are missed.

The objectives of this project are to:

1. Develop a predictive model for classifying defective parts.
2. Evaluate model performance under severe class imbalance using the custom FN & FP Score.
3. Identify key process variables associated with defective production.
4. Provide insights that can support manufacturing quality control and process improvement.

Together, these objectives outline a framework for using historical manufacturing data to improve defect detection.

2 Related Work

Machine learning has increasingly been applied to manufacturing systems for predictive quality monitoring and defect detection. Prior research shows that industrial process and test data can be used to learn relationships between operating conditions and product quality outcomes.

Zahrani et al. apply machine learning models to predict product quality and groove geometry in a circular laser-grooving manufacturing process. Their results show that data-driven models can capture nonlinear relationships between process parameters and final part characteristics, supporting predictive monitoring rather than purely reactive inspection [1].

More recent studies extend this direction using stronger ensemble methods for rare-defect prediction in industrial environments. Geng et al. propose a framework for predicting internal cracks in continuous steel casting using CatBoost combined with synthetic data generation, demonstrating improved detection of infrequent defect patterns in a real manufacturing context [2].

A central challenge in manufacturing defect prediction is severe class imbalance. Defective products may represent only a small fraction of observations, causing standard training and evaluation to favor the majority class and report high accuracy while missing rare failures. Kafunah et al. examine this issue in fault detection and evaluate imbalance-aware strategies such as oversampling and reweighting to improve minority-class detection, highlighting the importance of focusing on defect-class metrics such as precision, recall, and F1 rather than accuracy alone [3].

Cost-sensitive learning provides a complementary approach by explicitly assigning higher penalties to misclassifying defective samples, reflecting the higher operational risk of false negatives in manufacturing. Liu et al. introduce a cost-sensitive defect detection framework that modifies the training objective using an explicit cost structure to prioritize costly minority-class errors, and they report improved defect identification under imbalance [4].

Building on these findings, our project systematically compares interpretable and ensemble models, including logistic regression, decision trees, random forests, and boosting under multiple imbalance-mitigation strategies. We emphasize cost-sensitive random forests and minority-class performance metrics to evaluate model effectiveness for defect detection in a highly imbalanced manufacturing dataset.

3 Model and Method

3.1 Class Imbalance Strategies

The dataset comprises 229,999 test records with only 458 defective samples (0.2%). To address the imbalance, we evaluate four dataset manipulation techniques: Random Oversampling (duplicating minority samples), Random Undersampling (removing majority samples), SMOTE (synthetic sample generation), and Class-Weighted training (adjusting learning objectives to penalize minority-class errors). Each data sampling method was tested on every considered model type.

3.2 Considered Model Types

Logistic Regression

Logistic Regression provides high interpretability, modeling defect probability as a smooth function of sensor readings so engineers can quantify feature impact:

$$P(defective|x) = \sigma(\beta_0 + \beta^\top x) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top x)}} \tag{1}$$

Decision Trees

Decision Trees sort data through binary splits based on sensor thresholds. We use Gini Impurity, as it is numerically stable under extreme class imbalance:

$$Gini(S) = \sum_{k \in \{0,1\}} p_k(1 - p_k) \tag{2}$$

Splits are evaluated by their reduction in impurity:

$$\Delta = Gini(S) - \left(\frac{|S_L|}{|S|} Gini(S_L) + \frac{|S_R|}{|S|} Gini(S_R) \right) \tag{3}$$

Based on 5-Fold Cross Validation, the optimal maximum depth was set to 5 to prevent overfitting.

Random Forest

Random Forest aggregates multiple trees trained on bootstrap samples and random feature subsets, yielding a majority vote prediction:

$$\hat{y} = \text{mode}\{\hat{y}^{(1)}, \hat{y}^{(2)}, \ldots, \hat{y}^{(N)}\} \tag{4}$$

We evaluate both Balanced Random Forest and Cost-Sensitive Random Forest to address the severe class imbalance in the dataset. Balanced Random Forest uses internal undersampling for each tree, meaning each bootstrap sample is adjusted so that the defective class is better represented during training. Cost-Sensitive Random Forest keeps the original training distribution but applies class-weighted impurity, giving higher penalty to mistakes on defective samples. Comparing both approaches allows us to test whether defect detection improves more from changing the sample distribution or from changing the cost of minority-class errors.

Boosting (AdaBoost, XGBoost, LightGBM, CatBoost)

Boosting builds an ensemble by repeatedly reweighting samples to focus on previous mistakes. For our baseline, AdaBoost, the error is:

$$\epsilon_i = \sum_j w_j^i \delta(C_i(x_j) \neq y_j) \tag{5}$$

Classifier importance:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right) \tag{6}$$

Sample weights update:

$$w_j^{i+1} = \frac{w_j^i}{Z_i} \times \begin{cases} e^{-\alpha_i}, & C_i(x_j) = y_j \\ e^{\alpha_i}, & C_i(x_j) \neq y_j \end{cases} \tag{7}$$

We extend our analysis to advanced gradient boosting frameworks (XGBoost, LightGBM, and CatBoost) to gauge performance gains and processing efficiency against the AdaBoost baseline.

3.3 Data Split Comparison

Because the defective class represents only a very small portion of the dataset, model performance may be sensitive to how the data are divided into training and testing sets. A split with a larger training portion, such as 90/10, gives the model more examples to learn from, including more rare defective samples. However, it also leaves fewer defective samples in the test set, which can make metrics more sensitive to small changes in the number of false negatives. In contrast, a 70/30 split provides a larger test set and a more stable estimate of generalization performance, but it reduces the number of defective samples available during training.

To evaluate this trade-off, we compare three stratified train/test split ratios: 90/10, 80/20, and 70/30. Stratification is used so that each split preserves approximately the same defective/non-defective class proportion as the full dataset. This is especially important for this project because random splitting without stratification could produce a test set with too few defective examples, making recall, false negative rate, and the FN & FP Score unreliable.

The purpose of the split comparison is not only to identify the split with the best score, but also to determine whether model rankings remain stable across different evaluation conditions. If a model performs well only under one split ratio, its performance may be dependent on a favorable partition of rare defective samples. However, if a model remains competitive across multiple splits, this provides stronger evidence that it generalizes to unseen manufacturing data. For this reason, split sensitivity is treated as part of the model evaluation process rather than only a preprocessing decision.

3.4 Hyperparameter Tuning

Each model underwent hyperparameter tuning using 5-fold cross-validation, utilizing a customized penalty function that evaluated the trade-off between FN and FP weights. Table 1 summarizes the default and tuned hyperparameters across all models. Notably, the optimal hyperparameters remained consistent across all FN weight values (0.5–1.0), suggesting that the models' best configurations are robust regardless of how aggresive the scoring weight is penalizing rates of FN (FNR)

Table 1: Model Hyperparameters: Default vs. Tuned (Consistent Across All FN Weights)

Model	Parameter	Default	Tuned
Logistic Regression	C	1.0	0.1
	l1_ratio	0.5	1
Decision Tree	max_depth	5	5
	min_leaf	1	1
Balanced RF	max_depth	None	10
	n_estimators	100	100
Cost-Sensitive RF	max_depth	None	5
	n_estimators	100	300
XGBoost	max_depth	6	3
	learning_rate	0.3	0.05
	n_estimators	100	300
LightGBM	max_depth	-1	3
	learning_rate	0.1	0.05
	n_estimators	100	100
CatBoost	max_depth	6	6
	learning_rate	0.03	0.05
	iterations	1000	300

4 Experimental Design

4.1 Model Selection Criteria

We evaluate simple models alongside complex ensembles to consider the trade-off between interpretability and predictive performance. Our experimental phase primarily evaluates 120 unique scenarios via our custom "FN & FP Score" System, as well as secondary evaluation with other relevant metrics in the case that predictive ability (according to our custom scoring system) and interpretability do not significantly emphasize a best choice model. Our experimental phase evaluates 720 unique scenarios (incorporating 8 models, 5 sampling techniques, 3 train/test splits, and 6 custom weight profiles) via our custom "FN & FP Score" System. We also conduct secondary evaluations with other relevant metrics (such as PR-AUC and F1-minority score) in the event that predictive ability and interpretability do not significantly emphasize a clear best-choice model.

Conceptual model transparency and interpretability are also heavily considered for final model suggestion/recommendation.

4.2 "FN & FP Score" System for Evaluating Model Prediction Ability

In many classification tasks, reducing false negatives (FN) is often more critical than reducing false positives (FP). To systematically evaluate models with this consideration, we propose a weighted scoring system that incorporates both the false negative rate (FNR) and false positive rate (FPR).

4.2.1 Weighted Score Definition

We define the **"FN & FP Score"** for a model as a weighted sum of the FN Rate and FP Rate:

$$\text{FN \& FP Score} = (\text{FN weighting}) \times (\text{FN Rate}) + (\text{FP weighting}) \times (\text{FP Rate}) \tag{8}$$

where FN weighting and FP weighting are non-negative weights reflecting the relative importance of reducing false negatives versus false positives.

4.2.2 Auxiliary Equation

Assuming the weights are complementary, we can express the FP weighting in terms of the FN weighting:

$$\text{FP weighting} = 1 - \text{FN weighting} \tag{9}$$

4.2.3 Equivalent Equation

Substituting the complementary weighting into the score formula yields:

$$\text{FN \& FP Score} = (\text{FN weighting}) \times (\text{FN Rate}) + (1 - \text{FN weighting}) \times (\text{FP Rate}) \tag{10}$$

This equivalent equation shows that the score is a function of three components: FN weighting, FNR, and FPR.

4.2.4 Model Ranking Using the "FN & FP Score"

The "FN & FP" Score allows ranking models according to their performance while prioritizing reduction of false negatives. For a given FN weighting, the **lowest score** indicates the **best model** at that weighting. By varying FN weighting (e.g., 50%, 60%, ..., 100%), we can evaluate model performance under different prioritization schemes.

This approach provides a flexible and interpretable framework for model selection, emphasizing the trade-off between FN and FP according to domain-specific or safetyrequirements.

4.2.5 Subjectivity of FN Weighting

The FN weighting metric is subject to change based on organizational priorities. For example, incorporating the severe ethical or safety consequences of missing a defect could heavily increase the FN weighting. A completely unbiased, cost-based method for calculating the FN weighting is as follows:

$$\text{FN weighting} = \frac{\text{Cost of FN}}{\text{Cost of FN} + \text{Cost of FP}} \tag{11}$$

In this case, the respective costs could represent labor, material waste, system downtime, or reputational damage to the company.

4.3 Other Relevant Metrics

Due to the extreme class imbalance (99.8% non-defective to 0.2% defective), standard accuracy and ROC AUC can be highly misleading [5]. Instead, recall, precision, Area Under the Precision-Recall Curve (PR-AUC), and the F1-minority score are used as secondary metrics to evaluate our models. Summary visualizations of these metrics are embedded within our attached Jupyter Notebook. These metrics serve as a vital secondary evaluative criteria to ensure the models aren't achieving high recall simply by blindly flagging every item as a defect.

5 Results

5.1 Comparing Predictive Ability

To determine the best predictive models, we extracted the top-performing algorithms across our 6 distinct FN weighting profiles (0.5 through 1.0). The comprehensive ranking table can be referenced in the Appendix.

Analysis of the 720 evaluated scenarios yielded several consistent trends:

- **Dominance of CatBoost:** The **CatBoost** algorithm was unequivocally our best predictive model. Paired with a 90/10 train-test split and Random Undersampling, CatBoost secured the #1 rank across *every single weighting score*, consistently producing the lowest "FN & FP Score" of any combination.

- **AdaBoost and Logistic Regression as Strong Runners-Up:** As the penalty for False Negatives increased (weightings 0.6 to 1.0), **AdaBoost** firmly held the #2 overall spot. Interestingly, at a balanced 0.5 weighting, **Logistic Regression** (using Class-Weighting or Random Oversampling) overtook AdaBoost to claim the #2 and #3 ranks, making it highly competitive at lower FN penalties.
- **Optimal Data Treatment:** The **90/10 data split** was highly prevalent among the top-tier models, suggesting that maximizing the training data volume was crucial for this heavily imbalanced set. Additionally, **Random Undersampling** proved to be the most effective data manipulation strategy, utilized by almost all top-3 models across the higher weighting thresholds.

5.2 Comparing Interpretability

While predictive ability is paramount for defect detection, model interpretability dictates how easily engineers can trust, debug, and learn from the system.

Our #1 predictive model, **CatBoost** (alongside AdaBoost), is a complex gradient-boosting ensemble. It operates essentially as a "black box." While we can extract general feature importance to see *which* sensors matter most globally, we cannot easily see *how* the thresholds interact mathematically to produce a decision for a single component.

Conversely, **Logistic Regression**—which proved to be a remarkably strong runner-up, holding the #3 rank across the 0.7 to 1.0 weighting profiles and the #2 rank at 0.5—is a highly transparent linear model. It provides exact equations and odds ratios, allowing engineers to directly quantify how a 1-unit change in a specific sensor's reading shifts the mathematical probability of a defect.

Final Recommendation: If absolute maximum defect detection is required regardless of system opacity, **CatBoost (Random Undersampled, 90/10 split)** is the clear choice. However, if stakeholders require a transparent, explainable algorithm to guide manufacturing adjustments and understand system failures, **Logistic Regression** is a stellar alternative that sacrifices only a fractional margin of predictive ability for complete interpretability.

6 Conclusion and Future Work

6.1 Our Recommended Model

6.2 Limitations of Our Study

This study has several limitations that are relevant within the report. First, the dataset is extremely imbalanced, with only 458 defective samples out of nearly 230,000 observations. While various imbalance-handling strategies were applied, such a low defect rate makes it difficult for models to fully learn the characteristics of the minority class, potentially limiting generalization to new data.

The dataset also consists of numerical input features without detailed contextual information. While models such as Logistic Regression can provide coefficients, translating these into practical insights may require additional domain knowledge.

Additionally, the evaluation framework in terms of the cost-weighted error metric may reflect practical manufacturing priorities, but the choice of weighting can be subjective and may vary depending on management personnel bias.

Finally, the models were trained and evaluated on a static dataset and do not account for real-time production variability or changes in process conditions over time. As a result, model performance in a live manufacturing environment may differ from the results observed in this study.

6.3 Potential Future Improvements

Several improvements can be made for this study in future work. First, incorporating additional defective sample data would help address the extreme class imbalance and allow models to better learn minority class patterns. Collecting data across different operating conditions may also improve model generalization.

Also, the evaluation framework could be extended by exploring alternative cost-sensitive metrics or optimizing the weighting scheme based on specific manufacturing objectives. This would allow the model selection process to better reflect real-world priorities.

Finally, future work could involve deploying models in a real-time or streaming environment to evaluate performance under changing production conditions. This would provide insight into how models perform over time and whether periodic retraining or adaptation is necessary.

References

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Appendix

A. Detailed Scoring Results

Table 2: "FN & FP Scores" and Error Rates for Various Scenarios

Table 2: "FN & FP Scores" and Error Rates for Various Scenarios

FN Weight	Rank	Scenario	"FN & FP Score"	FNR	FPR
0.5	1	90/10 | Random Undersampling | CatBoost	0.034292137	0.045454545	0.023129728
0.5	2	90/10 | Random Oversampling | Logistic Regression	0.044256621	0.068181818	0.020331424
0.5	3	90/10 | Class-Weighted | Logistic Regression	0.044322207	0.068181818	0.020462595
0.5	4	80/20 | SMOTE | Logistic Regression	0.045513672	0.068181818	0.022845525
0.5	5	90/10 | SMOTE | Logistic Regression	0.045524603	0.068181818	0.022867387
0.5	6	90/10 | Random Undersampling | AdaBoost	0.046884502	0.045454545	0.048314459
0.5	7	90/10 | Random Undersampling | LightGBM	0.047186095	0.068181818	0.026190372
0.5	8	90/10 | Random Undersampling | XGBoost	0.047448436	0.068181818	0.026715054
0.5	9	80/20 | Random Undersampling | Logistic Regression	0.047538119	0.056818182	0.038258056
0.5	10	90/10 | Random Undersampling | Logistic Regression	0.047907533	0.068181818	0.027633247
0.6	1	90/10 | Random Undersampling | CatBoost	0.036524618	0.045454545	0.023129728
0.6	2	90/10 | Random Undersampling | AdaBoost	0.046598511	0.045454545	0.048314459
0.6	3	90/10 | Random Oversampling | Logistic Regression	0.04904166	0.068181818	0.020331424
0.6	4	90/10 | Class-Weighted | Logistic Regression	0.049094129	0.068181818	0.020462595
0.6	5	80/20 | Random Undersampling | Logistic Regression	0.049394132	0.056818182	0.038258056
0.6	6	80/20 | SMOTE | Logistic Regression	0.050047301	0.068181818	0.022845525
0.6	7	90/10 | SMOTE | Logistic Regression	0.050056046	0.068181818	0.022867387
0.6	8	90/10 | Random Undersampling | LightGBM	0.05138524	0.068181818	0.026190372
0.6	9	90/10 | Random Undersampling | XGBoost	0.051595112	0.068181818	0.026715054
0.6	10	90/10 | Random Undersampling | Logistic Regression	0.05196239	0.068181818	0.027633247
0.7	1	90/10 | Random Undersampling | CatBoost	0.0387571	0.045454545	0.023129728
0.7	2	90/10 | Random Undersampling | AdaBoost	0.046312519	0.045454545	0.048314459
0.7	3	80/20 | Random Undersampling | Logistic Regression	0.051250144	0.056818182	0.038258056
0.7	4	90/10 | Random Oversampling | Logistic Regression	0.0538267	0.068181818	0.020331424

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FN Weight	Rank	Scenario	"FN & FP Score"	FNR	FPR
0.7	5	90/10 | Class-Weighted | Logistic Regression	0.053866051	0.068181818	0.020462595
0.7	6	80/20 | SMOTE | Logistic Regression	0.05458093	0.068181818	0.022845525
0.7	7	90/10 | SMOTE | Logistic Regression	0.054587489	0.068181818	0.022867387
0.7	8	90/10 | Random Undersampling | LightGBM	0.055584384	0.068181818	0.026190372
0.7	9	90/10 | Random Undersampling | XGBoost	0.055741789	0.068181818	0.026715054
0.7	10	90/10 | Random Undersampling | Logistic Regression	0.056017247	0.068181818	0.027633247
0.8	1	90/10 | Random Undersampling | CatBoost	0.040989582	0.045454545	0.023129728
0.8	2	90/10 | Random Undersampling | AdaBoost	0.046026528	0.045454545	0.048314459
0.8	3	80/20 | Random Undersampling | Logistic Regression	0.053106157	0.056818182	0.038258056
0.8	4	90/10 | Random Oversampling | Logistic Regression	0.058611739	0.068181818	0.020331424
0.8	5	90/10 | Class-Weighted | Logistic Regression	0.058637973	0.068181818	0.020462595
0.8	6	80/20 | SMOTE | Logistic Regression	0.059114559	0.068181818	0.022845525
0.8	7	90/10 | SMOTE | Logistic Regression	0.059118932	0.068181818	0.022867387
0.8	8	90/10 | Random Undersampling | LightGBM	0.059783529	0.068181818	0.026190372
0.8	9	90/10 | Random Undersampling | XGBoost	0.059888465	0.068181818	0.026715054
0.8	10	90/10 | Random Undersampling | Logistic Regression	0.060072104	0.068181818	0.027633247
0.9	1	90/10 | Random Undersampling | CatBoost	0.043222063	0.045454545	0.023129728
0.9	2	90/10 | Random Undersampling | AdaBoost	0.045740536	0.045454545	0.048314459
0.9	3	80/20 | Random Undersampling | Logistic Regression	0.054962169	0.056818182	0.038258056
0.9	4	90/10 | Random Oversampling | Logistic Regression	0.063396779	0.068181818	0.020331424
0.9	5	90/10 | Class-Weighted | Logistic Regression	0.063409896	0.068181818	0.020462595
0.9	6	80/20 | SMOTE | Logistic Regression	0.063648189	0.068181818	0.022845525
0.9	7	90/10 | SMOTE | Logistic Regression	0.063650375	0.068181818	0.022867387
0.9	8	90/10 | Random Undersampling | LightGBM	0.063982673	0.068181818	0.026190372
0.9	9	90/10 | Random Undersampling | XGBoost	0.064035142	0.068181818	0.026715054
0.9	10	90/10 | Random Undersampling | Logistic Regression	0.064126961	0.068181818	0.027633247
1	1	90/10 | Random Undersampling | CatBoost	0.045454545	0.045454545	0.023129728
1	2	90/10 | Random Undersampling | AdaBoost	0.045454545	0.045454545	0.048314459

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FN Weight	Rank	Scenario	"FN & FP Score"	FNR	FPR
1	3	80/20 | Random Undersampling | Logistic Regression	0.056818182	0.056818182	0.038258056
1	4	80/20 | Random Undersampling | CatBoost	0.068181818	0.068181818	0.027720694
1	5	80/20 | Random Undersampling | LightGBM	0.068181818	0.068181818	0.030693892
1	6	90/10 | Random Oversampling | Logistic Regression	0.068181818	0.068181818	0.020331424
1	7	90/10 | Random Undersampling | LightGBM	0.068181818	0.068181818	0.026190372
1	8	90/10 | SMOTE | Logistic Regression	0.068181818	0.068181818	0.022867387
1	9	80/20 | SMOTE | Logistic Regression	0.068181818	0.068181818	0.022845525
1	10	90/10 | Class-Weighted | Logistic Regression	0.068181818	0.068181818	0.020462595

B. Bar Charts: Top 10 Models Per Weighting

The following bar charts include the top 10 scoring models for each False Negative weight (50%-100%), along with their "FN & FP Scores".

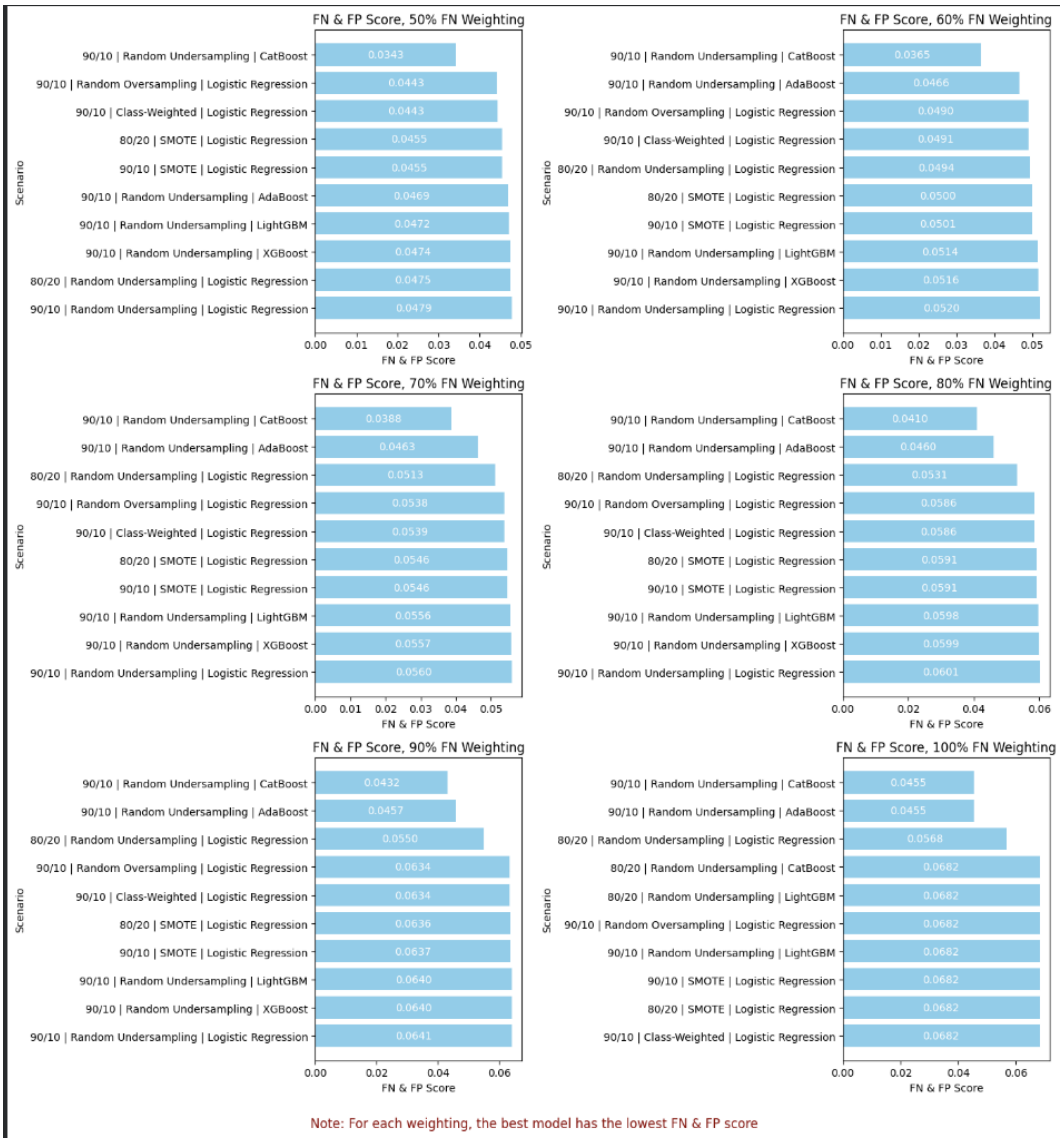

Figure 1: Model ranking results using the custom FN & FP Score.